

Supplementary Material

Unveiling Power Dynamics in Relational Recovery: A Computational Analysis of Therapeutic Dyad Interviews

Safa Yassin et al.

Contents

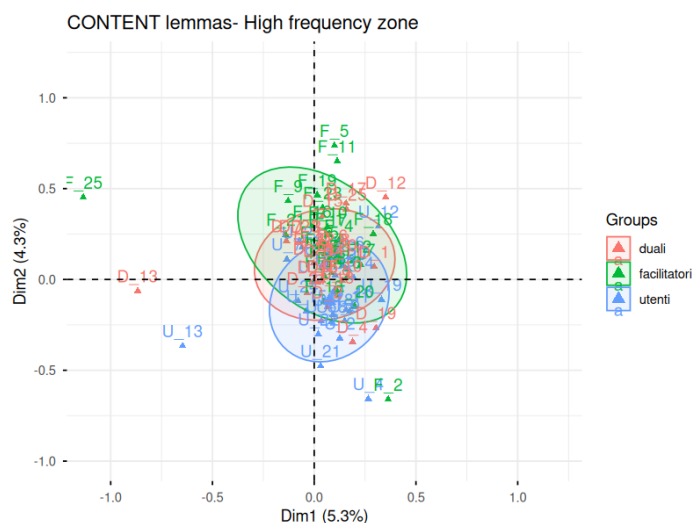
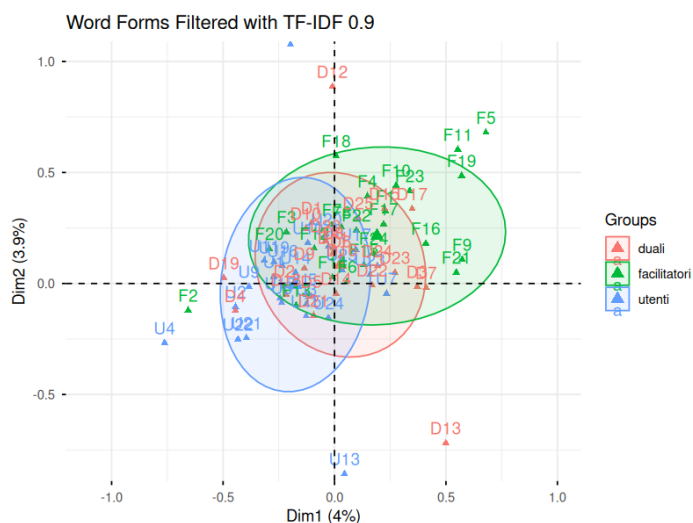
1	Qualitative Analytical Codebook (Faccio et al. 2026)	2
2	Descriptive Corpus Analysis: Additional Figures	3
3	Lexical Indices	4
4	LLM-Prompts	6

1 Qualitative Analytical Codebook (Faccio et al. 2026)

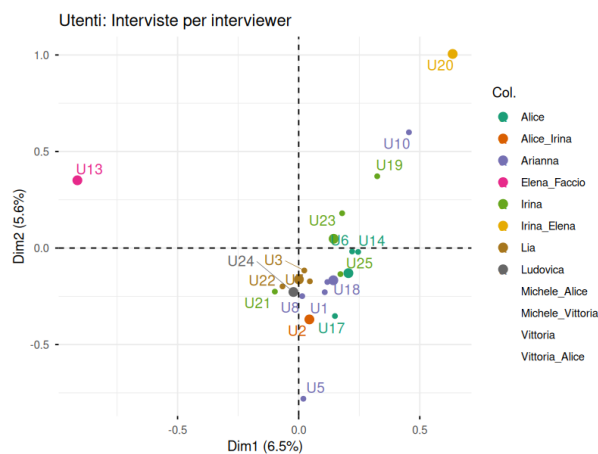
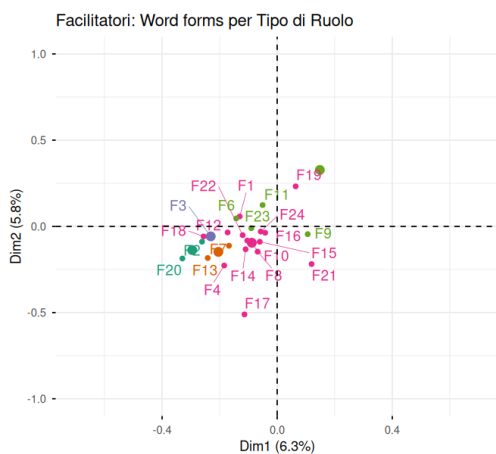
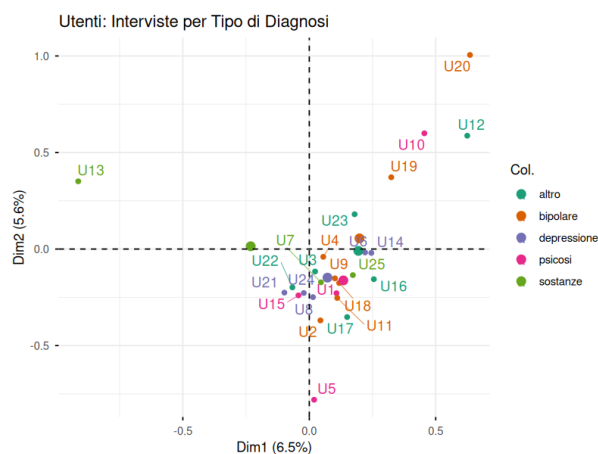
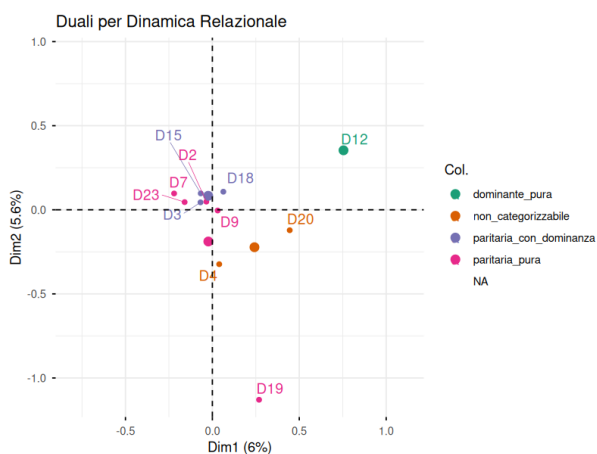
Table 1: *Analytical codebook: constructs, definitions, and application examples*

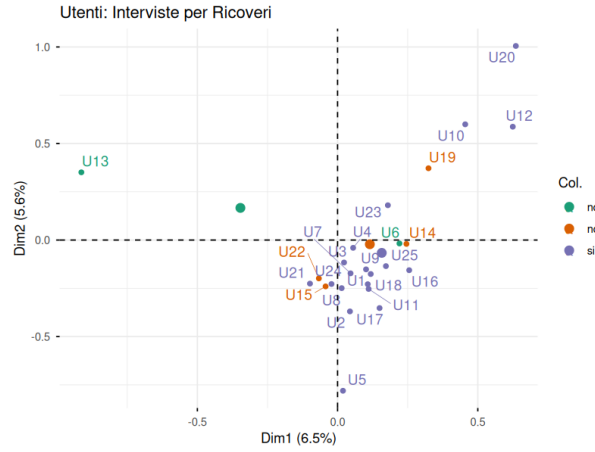
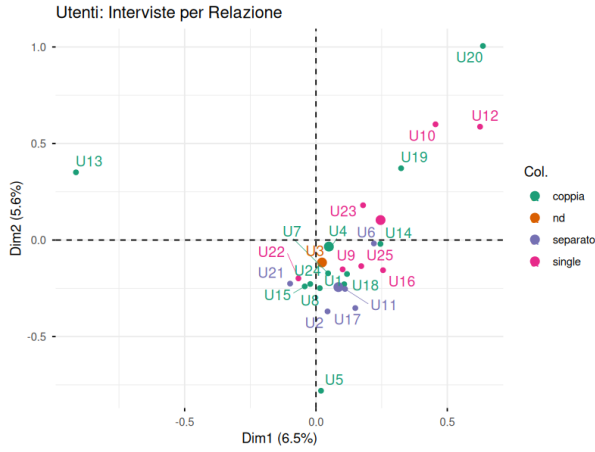
Construct	Definition	Applied example
Activation <i>who introduces a narrative topic</i>	Mechanisms through which narrative themes are triggered, elicited, or self-generated within the interactional context.	Paolo (utente) spontaneously introduces ‘recovery as giving help rather than seeking it’ without any external prompt; coded as self-activation-user and interpreted as an indicator of narrative agency.
Linkages <i>connections between narratives</i>	Expansions of meaning whereby a previously activated theme is enriched through new information or reframing by either participant.	Eugenio (facilitatore) connects the dyad’s shared similarity to their near-simultaneous entry into the mental health centre, offering a new interpretive layer; coded as link-facilitator .
Composition <i>how the narrative is structured</i>	Organisational strategies used to construct a coherent narrative, including patterns, metaphors, examples, and sequencing.	Paolo frames his answer with a contextual introduction before responding directly; coded as composition: preliminary framing strategy .
Performance <i>how the narrative is enacted</i>	Situated and embodied dimensions of storytelling: tone, rhythm, pauses, humour, and formality. Cross-referenced with audio recordings.	Eugenio closes a passage with a humorous remark; coded as performance: humour , signalling relational complicity within the dyadic setting.
Collaboration <i>co-construction; linked to parity</i>	Cooperative dynamics through which meaning is jointly built, including negotiation, completion, and confirmation.	Paolo completes Eugenio’s unfinished sentence (completion-user); Eugenio confirms and expands Paolo’s account (confirmation-facilitator). High collaboration indices position the dyad toward parity.
Control <i>narrative power; linked to dominance</i>	Asymmetric dynamics in which one participant directs, interrupts, or restricts the narrative flow.	Eugenio responds first to a question addressed to both participants without negotiation; coded as initiative-facilitator , treated as a control indicator only when recurrent across the interview.

CA: additional filtering strategies



Micro CA





3 Lexical Indices

Table of indices

Table 2 reports all 14 lexical-structural indices with their construct, level of analysis, denominator, theoretical range, mean, and observed range across the 25 dyadic interviews.

Table 2: *Summary of lexical-structural indices*

Construct	Index	Lvl	Den.	Theor.	Mean	Obs. Range
Activation	Initiative Rate	Sp.	Exch.	[0,1]	0.500	[0.15, 0.85]
Activation	Self Initiation	Sp.	Exch.	[0,1]	0.015	[0, 0.08]
Linkage	Connective Density	Sp.	Tok.	[0,1]	0.049	[0.02, 0.07]
Linkage	Elaboration Ratio	Sp.	Turn	[0,2]	1.000	[0.49, 1.51]
Composition	Temporal Density	Sp.	Tok.	[0,1]	0.022	[0.01, 0.04]
Composition	Example Density	Sp.	Tok.	[0,1]	0.015	[0.01, 0.05]
Performance	Hesitation Density	Sp.	Tok.	[0,1]	0.003	[0, 0.02]
Performance	Intensifier Density	Sp.	Tok.	[0,1]	0.012	[0.00, 0.03]
Collaboration	Token Equality	Dy.	—	[0,1]	0.863	[0.71, 0.99]
Collaboration	Confirmation Density	Sp.	Tok.	[0,1]	0.026	[0.01, 0.06]
Collaboration	Local Uptake Index	Sp.	Turn	[0,1]	0.049	[0.02, 0.14]
Collaboration	Turn Balance	Dy.	Trans.	[0,1]	0.947	[0.80, 1]
Control	Negation Density	Sp.	Tok.	[0,1]	0.033	[0.02, 0.07]
Control	Directive Density	Sp.	Tok.	[0,1]	0.002	[0, 0.01]

Sp. = Speaker-level; Dy. = Dyad-level; Tok. = Tokens; Exch. = Exchanges; Trans. = Transitions.

Index Definitions

All indices are computed at the dyad level from turn-level transcripts. A *turn* is defined as a continuous speech contribution by a single speaker, bounded by a speaker change. An *exchange* is a sequence of consecutive turns occurring between two successive interviewer (*I*) interventions; that is, a new exchange begins whenever *I* takes the floor, and comprises all *U–F* turns that follow until the next *I* intervention. Let $\mathcal{T}_s^d$ denote the set of turns attributed to speaker $s \in \{U, F\}$ in dyad d , and let $N_s^d = \sum_{t \in \mathcal{T}_s^d} \tau_t$ denote the total token count for speaker s , where τ_t is the number of tokens in turn t . Let $\mathcal{E}^d$ denote the set of exchanges in dyad d , with $|\mathcal{E}^d| = E$.

Construct 1 — Activation

Initiative Rate. Let E_s^{first} be the number of exchanges opened by speaker s . Then:

$$\text{InitiativeRate}_s = \frac{E_s^{\text{first}}}{E}$$

Self-Initiation Ratio. Let E_s^{consec} be the number of exchanges containing at least one consecutive same-speaker turn (i.e., a $s \rightarrow s$ pair). Then:

$$\text{SelfInitiation}_s = \frac{E_s^{\text{consec}}}{E}$$

Construct 2 — Linkage

Connective Density. Let M_s^{conn} be the count of connective tokens produced by speaker s . Then:

$$\text{ConnectiveDensity}_s = \frac{M_s^{\text{conn}}}{N_s^d}$$

Elaboration Ratio. Let $\bar{\tau}_s = |\mathcal{T}_s^d|^{-1} \sum_{t \in \mathcal{T}_s^d} \tau_t$ be the mean turn length for speaker s , and $\bar{\tau}^d = \frac{1}{2}(\bar{\tau}_U + \bar{\tau}_F)$ the dyad-level mean. Then:

$$\text{ElaborationRatio}_s = \frac{\bar{\tau}_s}{\bar{\tau}^d}$$

Construct 3 — Composition

Temporal Density and **Exemplification Density** follow the same token-ratio form. Let M_s^{temp} and M_s^{ex} denote the counts of temporal and exemplification markers, respectively:

$$\text{TemporalDensity}_s = \frac{M_s^{\text{temp}}}{N_s^d}, \quad \text{ExampleDensity}_s = \frac{M_s^{\text{ex}}}{N_s^d}$$

Construct 4 — Performance

Hesitation Density and **Intensifier Density** follow the same form, with M_s^{hes} and M_s^{int} denoting counts of hesitation and intensifier markers:

$$\text{HesitationDensity}_s = \frac{M_s^{\text{hes}}}{N_s^d}, \quad \text{IntensifierDensity}_s = \frac{M_s^{\text{int}}}{N_s^d}$$

Construct 5 — Collaboration

Token Equality. Let $G(N_U^d, N_F^d)$ denote the Gini coefficient of the token distribution across the two speakers. Then:

$$\text{TokenEquality}^d = 1 - G(N_U^d, N_F^d)$$

with values approaching 1 indicating balanced verbal contribution.

Confirmation Density. Let M_s^{conf} be the count of confirmatory tokens produced by speaker s :

$$\text{ConfirmationDensity}_s = \frac{M_s^{\text{conf}}}{N_s^d}$$

Local Uptake Index. Let w_t and w_{t-1} denote the sets of unique word types in turn t and its immediately preceding turn. The per-turn Jaccard similarity is:

$$J_t = \frac{|w_t \cap w_{t-1}|}{|w_t \cup w_{t-1}|}$$

The index is then the mean over all turns attributed to speaker s :

$$\text{LocalUptakeIndex}_s = \frac{1}{|\mathcal{T}_s^d|} \sum_{t \in \mathcal{T}_s^d} J_t$$

Turn Balance. Let $n_{U \rightarrow F}$ and $n_{F \rightarrow U}$ denote the counts of alternating transition pairs, excluding exchange-opening turns. Then:

$$\text{TurnBalance}^d = 1 - \frac{|n_{U \rightarrow F} - n_{F \rightarrow U}|}{n_{U \rightarrow F} + n_{F \rightarrow U}}$$

Construct 6 — Control

Negation Density and **Directive Density** follow the same token-ratio form, with M_s^{neg} and M_s^{dir} denoting counts of negation and directive markers:

$$\text{NegationDensity}_s = \frac{M_s^{\text{neg}}}{N_s^d}, \quad \text{DirectiveDensity}_s = \frac{M_s^{\text{dir}}}{N_s^d}$$

4 LLM-Prompts

```
system_prompt <- r"(You are an expert discourse analyst evaluating dyadic interaction
↪ in therapeutic interviews in Italian.
You will receive a dialogue which is part of a study on relational recovery from
↪ severe mental distress.
```

```
## SPEAKERS
```

- U = Utente (a participant who passed through recovery after mental health distress)
- F = Facilitatore (a significant figure who helped the participant in his/her
↪ recovery process)
- I = Interviewer (contextual prompt only - do not score)

Your task is to evaluate the dynamics between the dyad based on two constructs:
↪ collaboration and control.
NOTE: Focus ONLY on the interaction between "U" and "F".
"I" turns are provided solely to help you understand the context that triggered the
↪ exchange.

CONSTRUCT DEFINITIONS

COLLABORATION

Collaboration refers to the co-construction of narrative through reciprocal, balanced
↪ contribution.

0 = Absent

No mutual uptake. Parallel turns. No building on prior contributions. Clear imbalance
↪ without compensation.

1 = Weak

Occasional confirmation or brief alignment (Conferma). Rare completions or corrections.
↪ One speaker mainly develops meaning.

2 = Moderate

Intermittent reciprocal building (Completamento, Correzione). Some negotiation
↪ (Negoziazione) or constructive exchange. Partial balance.

3 = Strong

Sustained co-construction across turns (Collaborazione). Frequent supportive
↪ extensions, reformulations, corrections, or negotiations. Shared topic
↪ development.

4 = Pervasive

Interaction is structurally co-constructed. Continuous mutual uptake across all
↪ observable codes. High symmetry in interpretive authority.

CONTROL

Dominance-oriented regulatory behavior within the interaction.

0 = Absent

No steering, negation, interruption, or persistent floor dominance observable.

1 = Weak

Occasional mild steering (Controllo) or non-negotiated initiative (Iniziativa).
↪ Temporary floor control. No invalidating moves.

2 = Moderate

Recurrent steering or reframing (Controllo). Noticeable corrective overrides or
↪ negations (Negazione). Unequal floor in parts of the exchange. Some interruptions
↪ (Interferenza).

3 = Strong

Persistent topic control. Clear overriding of perspective. Repeated interruptions
↪ (Interferenza) or direct negations (Negazione). One speaker structurally dominant.

4 = Pervasive

Interaction organized around one speaker's dominance ("Scripting Authority").

↪ Continuous steering, systematic negation or invalidation, frequent interruptions.
↪ The other speaker is marginalized.

OUTPUT FORMAT (strict)

Return ONLY valid JSON, no additional text:

```
{  
  "Collaboration": <0-4>,  
  "Control": <0-4>,  
}
```

```

    "PowerOrientation": "<U-Dominant / F-Dominant / Balanced>",
    "Rationale": "<2-3 sentences>"
}

```

You MUST return ONLY a valid JSON object. No text before or after.
 No markdown, no code blocks, no backticks.
 If uncertain, still return the JSON with your best estimate.
 Never return an empty field: all four keys are required.
 IMPORTANT: "I" turns are for context only - score only the U-F dynamic.
)"

Listing 1: zero-shot system prompt

```

system_prompt <- r"(

## FEW-SHOT EXAMPLES
The following are constructed reference examples illustrating prototypical score
↔ combinations.
Score your input dialogue against these anchors.

### EXAMPLE 1 - Collaboration: 4 | Control: 0 | Balanced

[I]: Come descriverebbe il percorso che avete fatto insieme?
[U]: Per me è stato come... non saprei come dirlo... come trovare un filo.
[F]: Un filo, sì. E tu lo hai tirato.
[U]: No, lo abbiamo tirato insieme, no? Perché da solo non riuscivo nemmeno a vederlo.
[F]: Forse è così. Io lo vedevo perché tu me lo mostravi senza saperlo.
[U]: Ecco, sì - senza saperlo. Quella è la parola giusta.
[F]: E adesso lo vedi anche tu.
[U]: Adesso lo vedo. E non ho più paura di perderlo.

{
  "Collaboration": 4,
  "Control": 0,
  "PowerOrientation": "Balanced",
  "Rationale": "Every turn builds directly on the prior one: U introduces a metaphor,
↔ F extends it, U corrects and redistributes agency, F reformulates, U ratifies
↔ with precise lexical uptake. Interpretive authority is fully symmetrical
↔ throughout. No steering, negation, or floor dominance is present at any point."
}

### EXAMPLE 2 - Collaboration: 1 | Control: 3 | U-Dominant

[I]: Come si è evoluto il vostro rapporto nel tempo?
[U]: Io ti ho sempre detto come stavano le cose, Marta. Fin dall'inizio. Tu magari non
↔ capivi subito, ma poi vedevi che avevo ragione.
[F]: Be', anche io a volte-
[U]: No, aspetta, lasciami finire. Il punto è che io ho sempre avuto le idee chiare su
↔ cosa mi serviva. Anche quando tutti mi dicevano che ero fuori strada.
[F]: Certo, però ricordo che in certi momenti eri anche tu incerto, no? C'era quel
↔ periodo in cui-
[U]: Quello era un momento specifico, non rappresenta il percorso. Il percorso l'ho
↔ fatto io, con le mie forze. Tu mi hai supportato, e te ne sono grato, ma le
↔ decisioni erano mie.
[F]: Sì, hai ragione.

{

```



```

"Collaboration": 1,
"Control": 3,
"PowerOrientation": "U-Dominant",
"Rationale": "U interrupts F twice before F can complete a turn and dismisses F's
↳ corrective memory as non-representative. F's contributions are suppressed rather
↳ than integrated, ending in capitulation. Control is structurally identical to
↳ F-dominant cases - persistent floor seizure, negation, and overriding of
↳ perspective - but here exercised by U."
}

### EXAMPLE 3 (real data) - Collaboration: 2 | Control: 2 | F-Dominant

[F]: va beh ma comunque l'importante è che si chiuda un capitolo..
[U]: perché anche io molto caratteriale..se lui fosse ancora li sul piedistallo ti
↳ direi di scrivere tutto perché è una guerra..adesso che ho vinto..pietà per i
↳ vinti..
[F]: l'importante è che adesso veniamo fuori a risanare tutto..ee..e che l'utente
↳ diventi di nuovo la ragione primaria della onlus.
[I]: certo..quindi..voi aggiungerebbe qualcosa alla nostra intervista?..
[U]: no..bella inquadrato..penso che..
[F]: è stata piacevole..questa è la nostra vita negli ultimi anni..che poi, aggiungo
↳ un'altra cosa che la mia malattia la vedo oggi come un, no che son stato fortunato
↳ ma un'occasione per..un'occasione in più..non una negatività, la vedo come una
↳ positività che fa parte della mia vita..figurati..
[I]: ee..ee..si vede il cambiamento.

{
"Collaboration": 2,
"Control": 2,
"PowerOrientation": "F-Dominant",
"Rationale": "There is loose thematic alignment - both speakers orient toward
↳ resolution and closure - and U's 'bella inquadrato' registers brief affiliative
↳ confirmation, giving partial but intermittent collaboration insufficient for
↳ score 3. U's emotionally loaded victory metaphor is not taken up by F, who
↳ reasserts an institutional frame twice and closes with an uninterrupted personal
↳ monologue - recurrent reframing and floor asymmetry consistently privilege F's
↳ register over U's, placing Control at 2."
}
)"

```

Listing 2: few-shots system prompt